Outbreak of progressive inflammatory neuropathy following exposure to aerosolized porcine neural tissue.

In the fall of 2007, the Minnesota Department of Health was notified of 11 cases of an unexplained neurological illness, all linked to a pork processing plant, Quality Pork Processors, Inc., in Austin, MN. The cluster of workers had been experiencing similar symptoms, including fatigue, pain, numbness, and tingling in their extremities as well as weakness. The symptoms were described as more sensory than motor, and all patients had evidence of polyradiculoneuropathy with signs of nerve root irritation. An epidemiological investigation revealed that the only commonality between cases was their exposure to a pork brain extraction procedure involving compressed air. As relatives of the cases remained asymptomatic and all cultures for known pathogens were negative, the etiology of the syndrome seemed not to be infectious. Clinically, the syndrome was most akin to chronic inflammatory demyelinating polyneuropathy. Laboratory tests corroborated the clinical findings, revealing inflammation of peripheral nerves and nerve roots; however, these cases also had features clinically distinct from chronic inflammatory demyelinating polyneuropathy as well as laboratory testing revealing a novel immunoglobulin G immunostaining pattern. This suggested that the observed inflammation was the result of 1 or more unidentified antigens. This syndrome was ultimately dubbed progressive inflammatory neuropathy and was theorized to be an autoimmune reaction to aerosolized porcine neural tissue. Since the investigation's outset, 18 cases of progressive inflammatory neuropathy have been identified at the Minnesota pork processing plant, with 5 similar cases at an Indiana plant and 1 case at a Nebraskan plant. The plants in which cases have been identified have since stopped the use of compressed air in removing pork brains. All cases have stabilized or improved, with some requiring immunosuppressive and analgesic treatment. The study of progressive inflammatory neuropathy is ongoing, and the details of this investigation highlight the value of epidemiological principles in the identification and containment of outbreaks while researchers attempt to uncover the unique pathophysiology and potential etiology of the illness. Mt Sinai J Med 76:442-447, 2009.